CACS402: Cloud computing
BCA 7th Semester

Mechi Multiple Campus

Bhadrapur

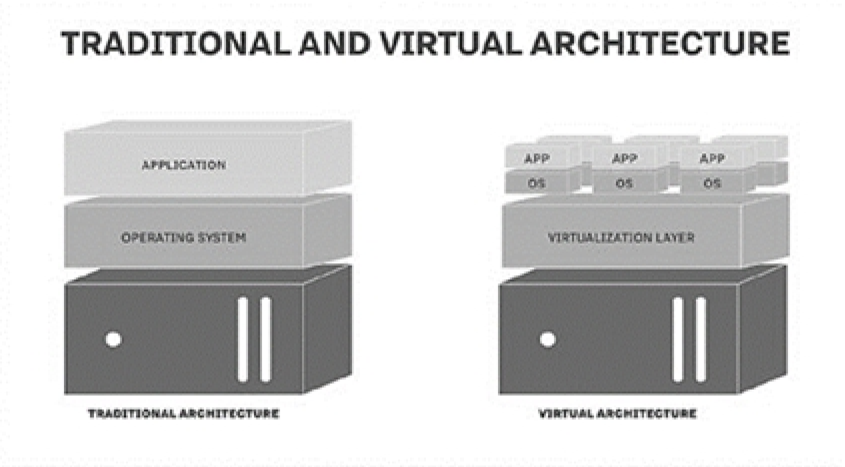
Unit 3: Cloud Virtualization technology

- Overview of Virtualization technique
- Types of Virtualization
- Implementation Levels of Virtualization Structures
- Virtualization benefits
- Server virtualization
- Hypervisor management software
- Virtual infrastructure requirements

OVERVIEW OF VIRTUALIZATION

TRADITIONAL & VIRTUAL ENVIRONMENT

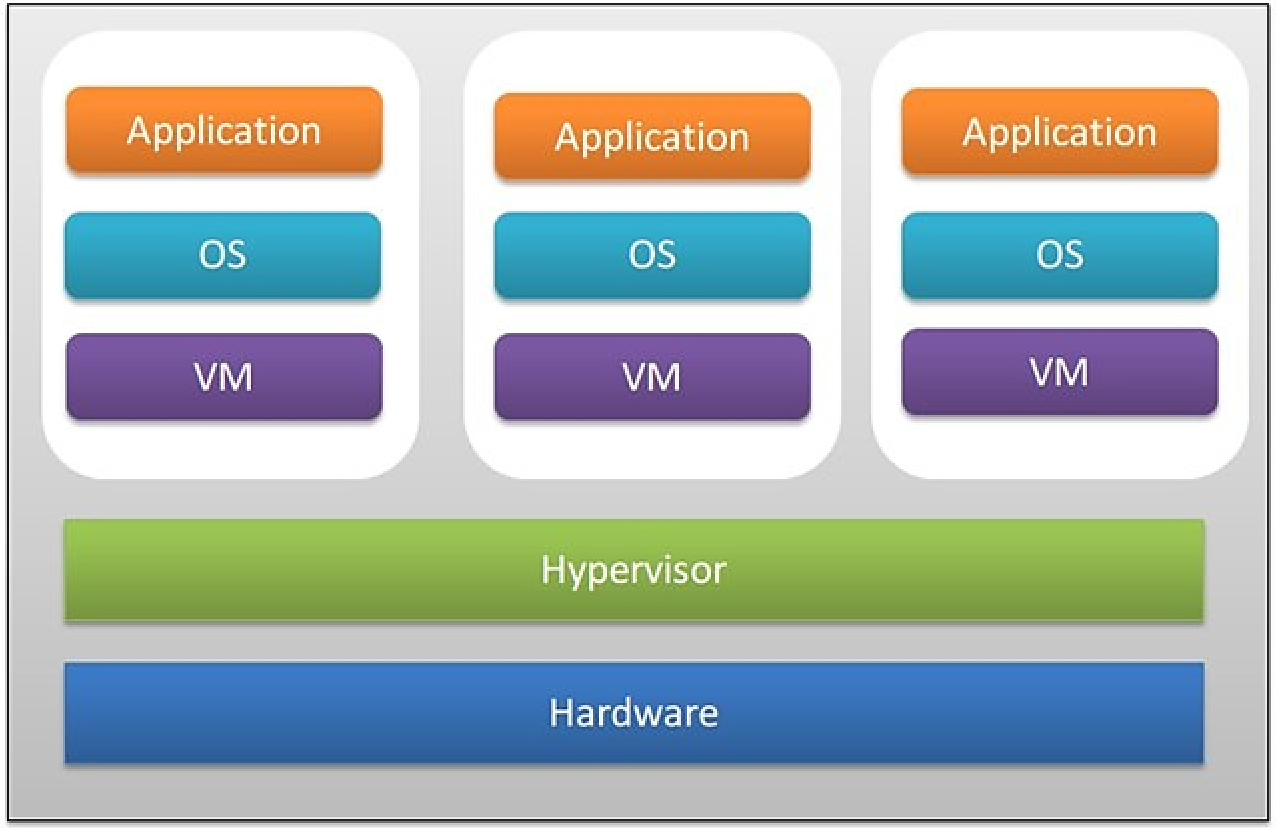
- In a typical physical computing environment, software like an operating system (OS) or corporate application has direct access to the underlying computer hardware and components, such as the CPU, memory, storage, specific chipsets, and OS driver versions.
- This caused significant problems with program setup and made it impossible to relocate or reinstall the software on new hardware, such as when restoring backups after a breakdown or disaster.
- Virtualization involves the installation of a hypervisor, which acts as an intermediary between the program and the underlying hardware. Once a hypervisor is installed, the software uses virtual representations of computer components, such as virtual processors, rather than actual processors.



Overview of Virtualization technique

- Virtualized computing resources are supplied as separate instances known as Virtual Machines (VMs), into which operating systems and applications can be loaded.
- Virtualized systems may host numerous VMs at the same time, but each VM is logically separated from the others. This implies that a malware attack or a VM crash will not impact the other VMs. Support for numerous VMs significantly improves system usage and efficiency.
- Instead of purchasing ten different servers to host ten physical applications, a single virtualized server might support the same ten applications deployed on ten virtual machines on the same system.
- This increased hardware utilization is a significant benefit of virtualization, as it provides great potential for system consolidation, decreasing the number of servers and power consumption in business data centers.
- The formation of a virtual version of something, such as a server, a desktop, a storage device, an operating system, or network resources is referred to as virtualization.
- Virtualization is a technology that allows numerous consumers and organizations to share a single physical instance of a resource or application. It accomplishes this by giving a logical name to a physical storage device and delivering a reference to that physical resource when needed.
- Virtualization can be applied very broadly to just about everything such as: Memory ,Networks ,Storage ,Hardware, Operating systems, Applications

Concept of Virtualization



Concept of Virtualization

- Usually, business data centers include a large number of servers, the majority of which are inactive since the responsibility is fulfilled by only a few servers on the network. This wastes expensive resources like hardware, electricity, maintenance, and cooling requirements.
- By dividing a real server into numerous virtual servers, virtualization helps to enhance resource utilization. These virtual servers seem and behave as if they were independent physical servers, each with its operating system and applications.
- Virtualization is a concept that is utilized in practically every IT architecture to assist double the capacity of physical devices. It aids in making the most use of current resources, lowering total corporate costs.
- Virtualization software (hypervisor) is used by businesses to construct virtual computers, networks, desktops, and servers.
- Virtualization software simplifies IT management and makes it more cost-effective to buy and run. It brings about a variety of favorable changes, such as lower hardware costs, improved disaster recovery solutions, higher IT agility, improved performance, and faster resource availability.
- The hypervisor enables virtual computers to run independently of the underlying physical hardware. A virtual machine, for example, can be relocated from one physical host to another, or its virtual disks can be switched from one kind of storage to another, without impacting the virtual machine's operation.
- Because virtual machines are detached from the underlying physical hardware, virtualization enables you to pool physical computing resources like CPUs, memory, storage and networking into pools of resources that can be made dynamically and flexibly accessible to virtual machines.

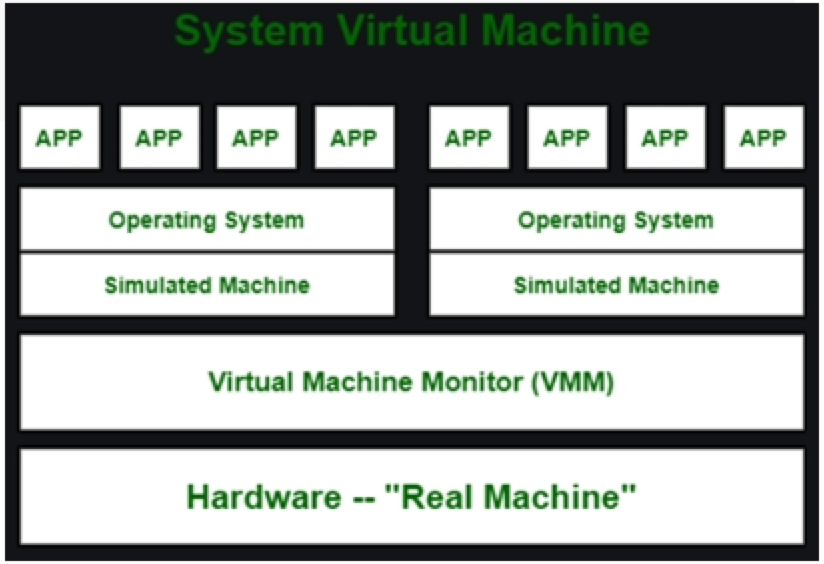
VIRTUAL MACHINE

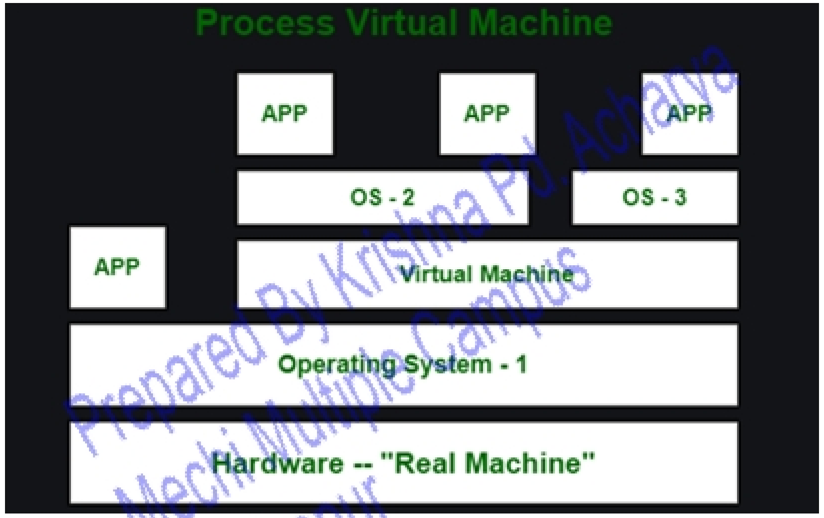
- A virtual machine is a computer program that, like a physical computer, runs an operating system and applications. The virtual machine is made up of a set of specification and configuration files and is supported by a host's physical resources. Every virtual machine has virtual devices that perform the same functions as actual hardware while providing extra mobility, management, and security benefits.
- A virtual machine (VM) is a software program that not only behaves like a separate computer but can also do functions such as executing apps and programs like a separate computer. A virtual computer sometimes referred to as a "guest," is constructed within another computer environment known as a "host." A single host may support several virtual machines at the same time.
- ***Examples:***
- VirtualBox (Windows/Mac/Linux, Free)
- Parallels (Windows/Mac/Linux, $79.99*)
- VMware (Windows/Linux, Basic: Free, Premium: $189*)
- QEMU (Linux, Free)
- Windows Virtual PC (Windows, Free)

System vs Process Virtual Machine

» 1. **System Virtual Machine**: These types of virtual machines gives us complete system platform and gives the execution of the complete virtual operating system. Just like virtual box, system virtual machine is providing an environment for an OS to be installed completely. We can see in below image that our hardware of Real Machine is being distributed between two simulated operating systems by Virtual machine monitor. And then some programs, processes are going on in that distributed hardware of simulated machines separately.

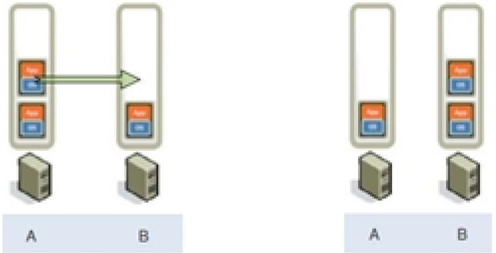
» 2.**Process Virtual Machine** : While process virtual machines, unlike system virtual machine, does not provide us with the facility to install the virtual operating system completely. Rather it creates virtual environment of that OS while using some app or program and this environment will be destroyed as soon as we exit from that app. Like in below image, there are some apps running on main OS as well some virtual machines are created to run other apps. This shows that as those programs required different OS, process virtual machine provided them with that for the time being those programs are running. Example – Wine software in Linux helps to run Windows applications



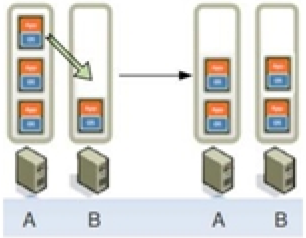


Benefits of using Virtual Machines

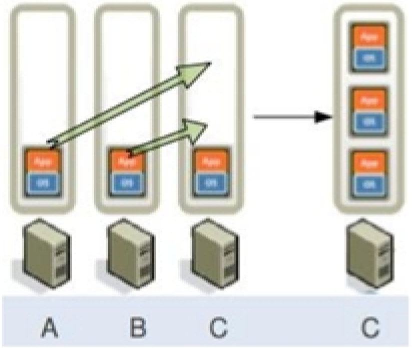
» VM Migration: Task of moving a virtual machine from one physical hardware environment to another one.

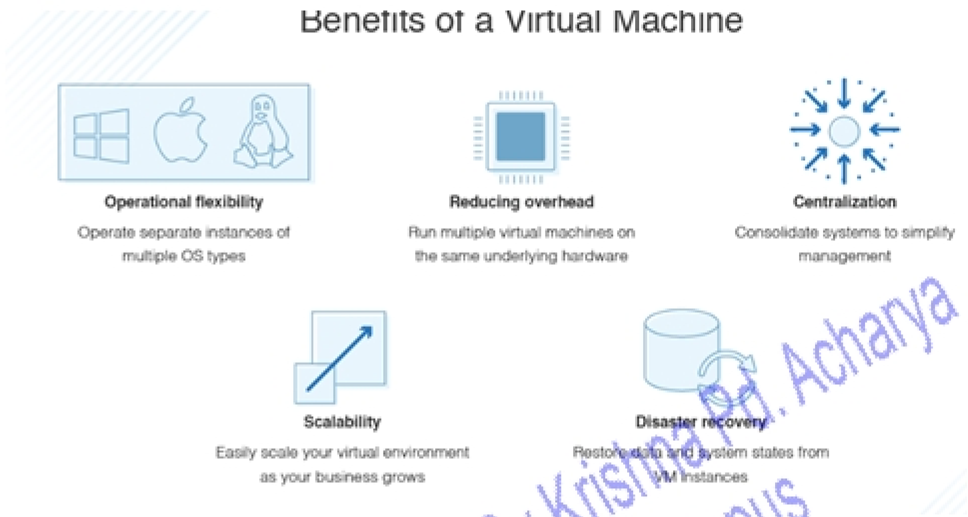


» Load balancing allows better response time.



» Consolidation: Reduces number of Physical Machine requirement.



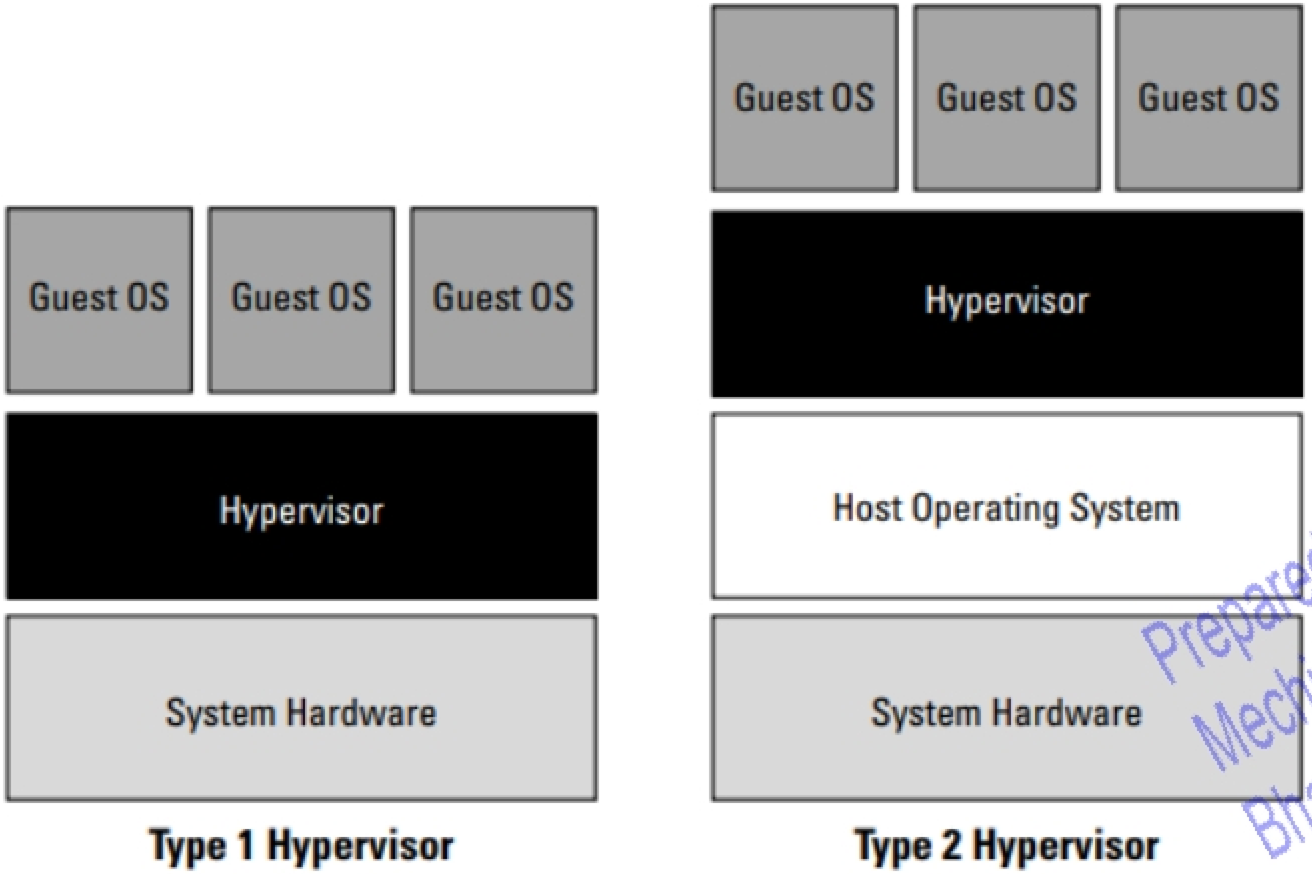


Hypervisor

- A hypervisor is software, hardware or a firmware that provides virtual partitioning capabilities which runs directly on hardware.
- Also known as VMM(Virtual Machine Monitor)
- It is defined as the virtual machine manager which allows multiple operating systems to run on a system at a time providing resources to each OS without any interaction.
- Hypervisor controls all the guest systems.
- As the operating system number increases managing is difficult which leads to security issues.
- If a hacker gets control over the hypervisor he can control the guest systems by knowing the behavior of the system which causes data processing damage.
- Advanced protection system is to be developed to monitor the activities of the guest Virtual machine.

Hypervisor

- **Type 1 Hypervisor –** Bare metal hypervisor (VMware ESXi)
- **Type 2 Hypervisor –** Hosted hypervisor (VMware Workstation)



Type 1 VM or native VM

- A low-level program is required to provide system resource access to virtual machines, and this program is referred to as the hypervisor or Virtual Machine Monitor (VMM).
- A hypervisor running on bare metal is a Type 1 VM or native VM.
- Examples of Type 1 Virtual Machine Monitors are LynxSecure, RTS Hypervisor, Oracle VM, Sun xVM Server, VirtualLogix VLX, VMware ESX and ESXi, and Wind River VxWorks, among others.
- The operating system loaded into a virtual machine is referred to as the guest operating system, and there is no constraint on running the same guest on multiple VMs on a physical system.
- Type 1 VMs have no host operating system because they are installed on a bare system.
- An operating system running on a Type 1 VM is a full virtualization because it is a complete simulation of the hardware that it is running on.

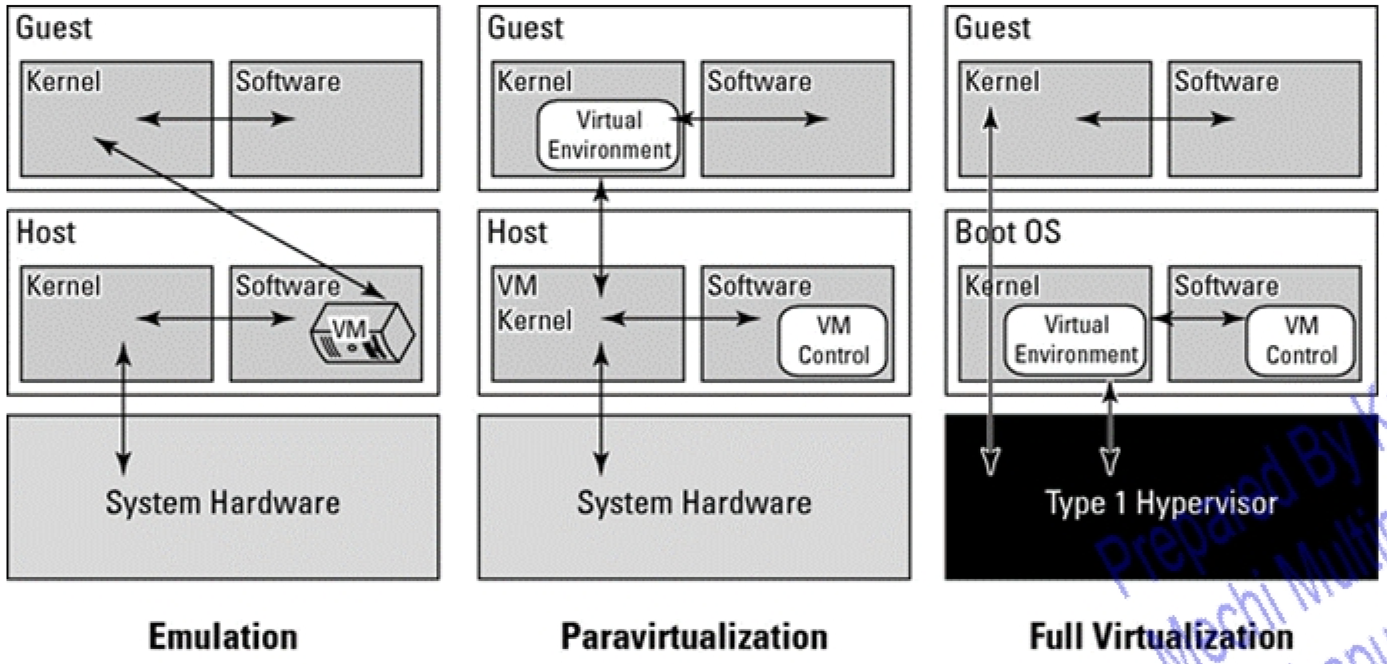
Type 2 or hosted VM

- Some hypervisors are installed over an operating system and are referred to as Type 2 or hosted VM.
- Examples of Type 2 Virtual Machine Monitors are Containers, KVM, Microsoft Hyper V, Parallels Desktop for Mac, Wind River Simics, VMWare Fusion, Virtual Server 2005 R2, Xen, Windows Virtual PC, and VMware Workstation 6.0 and Server, among others.
- On a Type 2 VM, a software interface is created that emulates the devices with which a system would normally interact.
- This abstraction is meant to place many I/O operations outside the virtual environment, which makes it both programmatically easier and more efficient to execute device I/O than it would be inside a virtual environment.
- This type of virtualization is sometimes referred to as paravirtualization, and it is found in hypervisors such as Microsoft's Hyper-V and Xen.

Types of virtualization based on hardware implementation

» Virtualization techniques can be applied at different layers in a computer stack: hardware layer (including resources such as the computer architecture, storage, network, etc.), the operating system layer and application layer. Examples of **virtualization types** are:

- Emulation (EM)
- Paravirtualization (PV)
- Full-Virtualization (FV)



Emulation , Paravirtualization and Full/native Virtualization

Emulation

- VM emulates/simulates complete hardware
- Unmodified guest OS for a different PC can be run Example: Bochs, VirtualPC for Mac, QEMU
- In emulation, the virtual machine simulates complete hardware, so it can be independent of the underlying system hardware.
- A guest operating system using emulation does not need to be modified in any way.

Paravirtualization

- VM does not simulate hardware
- Use special API that a modified guest OS must use example Xen, VMWare ESX Server
- Paravirtualization requires that the host operating system provide a virtual machine interface for the guest operating system and that the guest access hardware through that host VM.
- An operating system running as a guest on a paravirtualization system must be ported to work with the host interface.

Emulation , Paravirtualization and Full/native Virtualization

Full/native Virtualization

- Full/native Virtualization VM simulates "enough" hardware to allow an unmodified guest OS to be run in isolation
- Example IBM VM family, VMWare Workstation, Parallels
- In a full virtualization scheme, the VM is installed as a Type 1 Hypervisor directly onto the hardware.
- All operating systems in full virtualization communicate directly with the VM hypervisor, so guest operating systems do not require any modification.
- Guest operating systems in full virtualization systems are generally faster than other virtualization schemes.

Types of virtualization based on resources

Server Virtualization

- Server Virtualization, also known **as hardware-based virtualization**, divides a **physical server into multiple** virtual servers that run guest operating systems independently.
- According to **Market Research Future analysis**, Server Virtualization market is expected to grow up to USD 10,000 Million by 2026.
- To setup virtual servers, the server administrators utilize a software application such as a hypervisor to divide one physical server into numerous isolated virtual environments.
- Each VM is an independent server, even though it is running on just a part of the actual underlying computer hardware. This aspect enables businesses to efficiently utilize physical computer hardware and realize a greater return on their hardware investments.
- Moreover, the server virtualization arrangement helps a company divide its operations to manage complex workflows. For example, a team of graphics personnel can work on **a Mac OS**, whereas a team of accounting professionals will use a **Windows environment and** this diversified usage won't be an impediment.
- Server virtualization software reduces the need for multiple physical servers and requires a lesser amount of hardware. Thus, it's a highly effective cost reduction technology.
- Server virtualization has a major cost benefit: it allows you to **consolidate a large number** of physical machines onto fewer physical machines that host the virtual machines. This increase in **computing efficiency** results in **lower space**, **maintenance, cooling, and electricity costs,** besides the obvious reduction in procurement costs for the machines.
- An additional benefit is that fewer physical machines and lower electricity bills translate to environment friendliness.

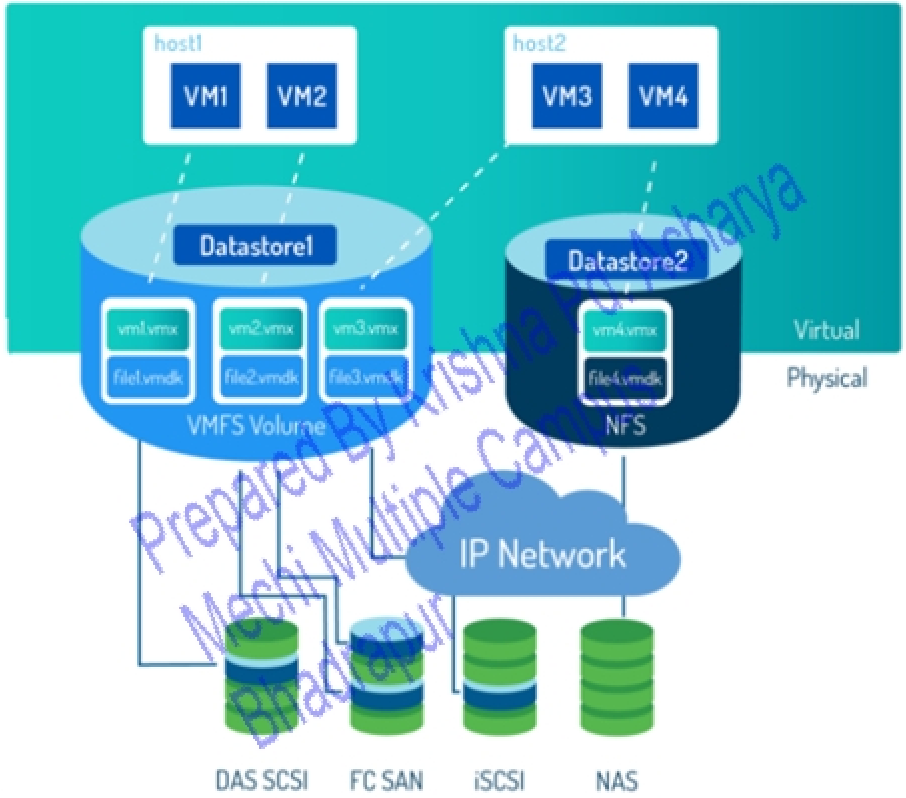
Types of virtualization based on resources

Server Virtualization

- **Types of Server virtualization**
  - Full Virtualization
  - Paravirtualization
  - Operating System virtualization
- **Advantages of Server Virtualization**
  - Cost Saving
  - Increases uptime
  - Improves Efficiency
  - Improves Disaster Recovery
  - Image-based backups
  - Allows to move to the cloud
- **Disadvantages of Server Virtualization**
  - Security Risk
  - Not all applications can be virtualized
  - Upfront costs

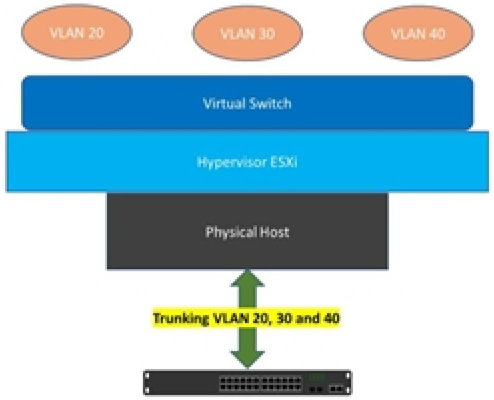
Types of virtualization based on resources

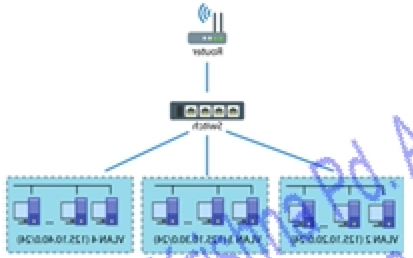
- **Storage Virtualization**
- Storage virtualization performs resource abstraction in a way that the **multiple physical storage arrays are virtualized as a single storage pool** with direct and independent access.
- The storage virtualization software aggregates and manages storage in various storage arrays and serves it to applications whenever needed.
- The centralized virtual storage increases flexibility and availability of resources needed. This data virtualization and centralization is easily manageable from a central console. It allows users to manage and access multiple arrays as a single storage unit.
- Storage capacity is pooled and distributed to the VMs
- Physical storage devices are partitioned into logical storage
- How do VMs access data center storage?
- VMs are stored as VMDK **(.vmdk) files** on datastore
- VM configuration files (VM settings) are stored as VMX (.vmx) files
- Top storage virtualization platforms include:
  - FalconStor Network Storage Server
  - IBM Storage Virtualization
  - QUADStor Systems



Types of virtualization based on resources

- **Network Virtualization**
- In network virtualization, all physical networking tools and other resources are **integrated into one software-based resource**. It further breaks the bandwidth into multiple channels, independent of one another.
- A network virtualization software **creates a tunnel through** the existing network. This way, the admins can run VMs without the need for individual network reconfiguration.
- The bandwidth division thus allows users to assign the channels to multiple systems in real-time. Three of the best network virtualization tools include:
  - 6WIND Virtual Accelerator
  - Cisco Nexus Virtual Services Appliance
  - JunosV App Engine





- Physical components that make up the physical network are virtualized to create a virtual network
- vSwitch: Virtual switch that virtual devices can connect to in order to communicate with each other
- vLAN: Virtual Local Area Network that is segmented into groups of ports isolated from one another, creating different network segments

Types of virtualization based on resources

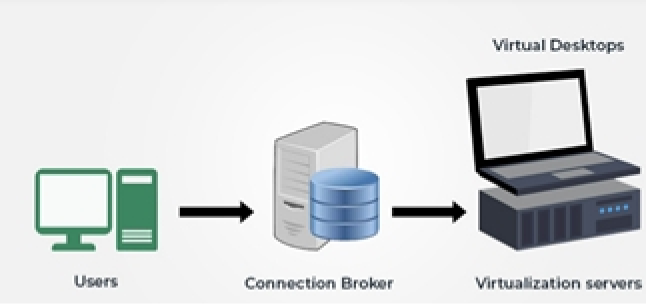
Desktop Virtualization

» The desktop Virtualization approach is also known as a client-server computing model. It stores any desktop environment virtually on a centralized remote server.

» Desktop virtualization software helps with workstation load and offers; smooth employee onboarding, redundant process removal, disaster recovery, single interface customization, and more.

» Top desktop virtualization software platforms include:

- Amazon WorkSpaces
- Citrix XenDesktop
- VMware Workstation and VMware Fusion



» It allows the user's OS to be remotely stored on a server in the data center and allows the users to access their desktop virtually, from any location through a different machine.

» Designed to solve computing resource issues faced by the mobile workforce (workers that need computing without the hardware)

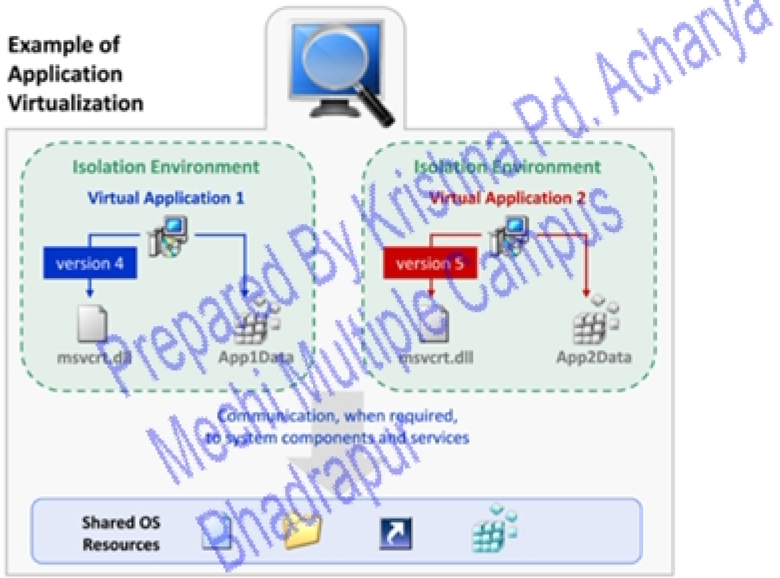
» VMware Horizon takes the resources needed to create a desktop environment from data centers and delivers it to the end-user's device

» Main benefits include user mobility, portability, easy management of software installation, updates and patches.

Types of virtualization based on resources

Application Virtualization

- Application virtualization hosts individual applications in which a virtual environment is entirely separate from the underlying operating system (OS).
- Application virtualization makes it possible to encapsulate all the elements of any particular application and run them independently from the underlying OS.
- It means that the machine that is running the application does not have that application installed. It resides on a (virtual machine) VM in a remotely located server.
- Some of the top platforms that specialize in application virtualization software are:
  - Citrix Virtual Apps and Desktops
  - Microsoft App-V
  - Parallels Remote Application Server (RAS)



Implementation Levels of Virtualization Structures

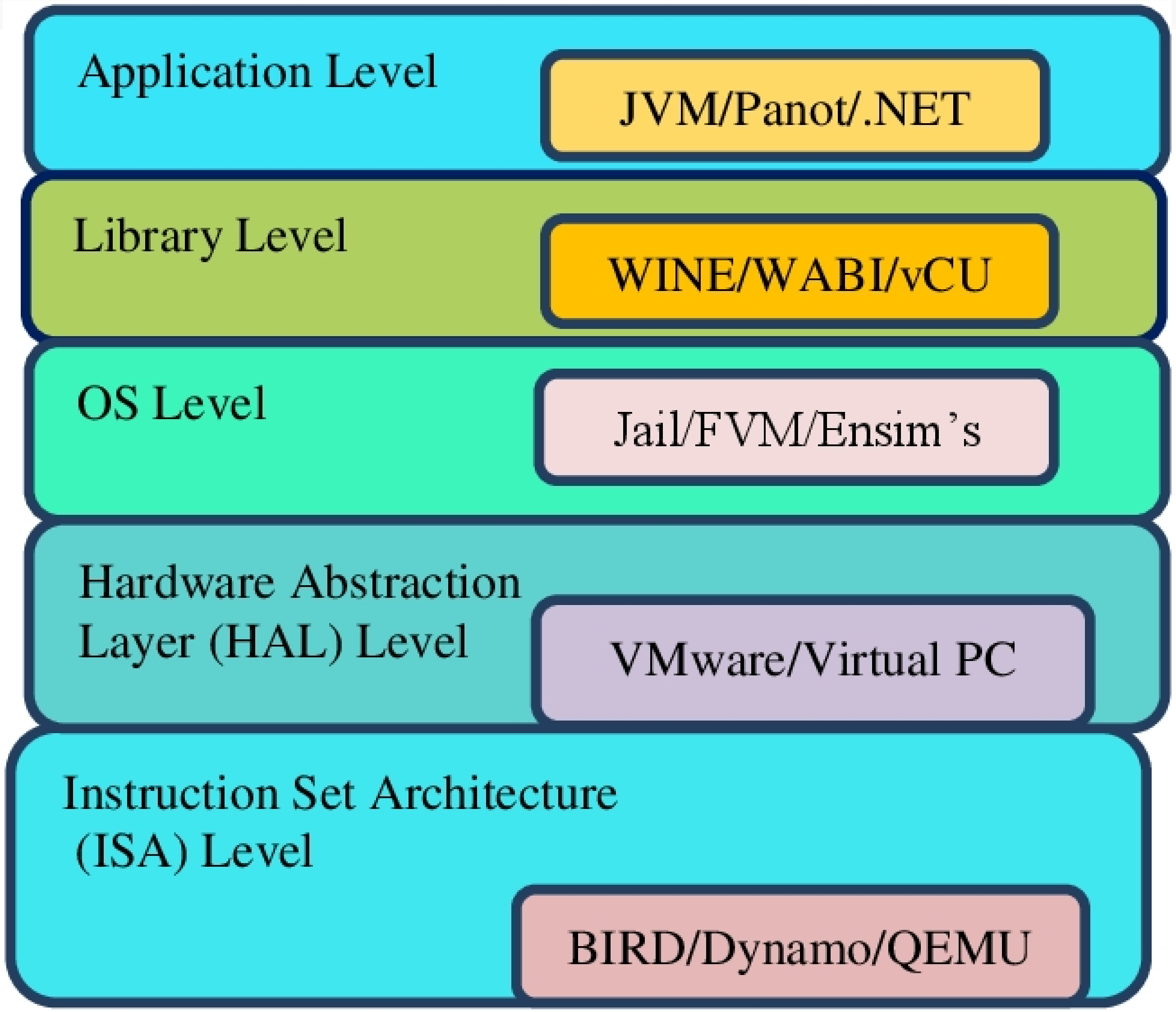


Fig. 2: Implementation Levels of Virtualization.

Implementation Levels of Virtualization Structures

Instruction Set Architecture Level (ISA)

- ISA virtualization can work through ISA emulation. This is used to run many legacy codes that were written for a different configuration of hardware.
- These codes run on any virtual machine using the ISA. With this, a binary code that originally needed some additional layers to run is now capable of running on the x86 machines. It can also be tweaked to run on the x64 machine.
- For the basic emulation, an interpreter is needed, which interprets the source code and then converts it into a hardware format that can be read. This then allows processing.

Hardware Abstraction Level (HAL)

- True to its name HAL lets the virtualization perform at the level of the hardware. This makes use of a hypervisor which is used for functioning. At this level, the virtual machine is formed, and this manages the hardware using the process of virtualization. It allows the virtualization of each of the hardware components, which could be the input-output device, the memory, the processor, etc.
- Multiple users will not be able to use the same hardware and also use multiple virtualization instances at the very same time. This is mostly used in the cloud-based infrastructure.

Implementation Levels of Virtualization Structures

Operating System Level

- At the level of the operating system, the virtualization model is capable of creating a layer that is abstract between the operating system and the application. This is an isolated container that is on the operating system and the physical server, which makes use of the software and hardware. Each of these then functions in the form of a server.
- When there are several users, and no one wants to share the hardware, then this is where the virtualization level is used. Every user will get his virtual environment using a virtual hardware resource that is dedicated. In this way, there is no question of any conflict.

Library Level

- The operating system is cumbersome, and this is when the applications make use of the API that is from the libraries at a user level. These APIs are documented well, and this is why the library virtualization level is preferred in these scenarios. API hooks make it possible as it controls the link of communication from the application to the system.

Application Level

- The application-level virtualization is used when there is a desire to virtualize only one application and is the last of the implementation levels of virtualization in cloud computing. One does not need to virtualize the entire environment of the platform.
- This is generally used when you run virtual machines that use high-level languages. The application will sit above the virtualization layer, which in turn sits on the application program.
- It lets the high-level language programs compiled to be used in the application level of the virtual machine run seamlessly.

Virtual infrastructure architecture

- A virtual infrastructure architecture can help organizations transform and manage their IT system infrastructure through virtualization. But it requires the right building blocks to deliver results. These include:
- **Host**: A virtualization layer that manages resources and other services for virtual machines. Virtual machines run on these individual hosts, which continuously perform monitoring and management activities in the background. Multiple hosts can be grouped together to work on the same network and storage subsystems, culminating in combined computing and memory resources to form a cluster. Machines can be dynamically added or removed from a cluster.
- **Hypervisor**: A software layer that enables one host computer to simultaneously support multiple virtual operating systems, also known as virtual machines. By sharing the same physical computing resources, such as memory, processing and storage, Cloud Computing the hypervisor stretches available resources and improves IT flexibility.
- **Virtual machine**: These software-defined computers encompass operating systems, software programs and documents. Managed by a virtual infrastructure, each virtual machine has its own operating system called a guest operating system. The key advantage of virtual machines is that IT teams can provision them faster and more easily than physical machines without the need for hardware procurement. Better yet, IT teams can easily deploy and suspend a virtual machine, and control access privileges, for greater security. These privileges are based on policies set by a system administrator.
- **User interface**: This front-end element means administrators can view and manage virtual infrastructure components by connecting directly to the server host or through a browser based interface.

Virtual infrastructure requirements

» Virtual infrastructure requirements From design to disaster recovery, there are certain virtual infrastructure requirements organizations must meet to reap long-term value from their investment.

» **Plan ahead**: When designing a virtual infrastructure, IT teams should consider how business growth, market fluctuations and advancements in technology might impact their hardware requirements and reliance on compute, networking and storage resources.

» **Look for ways to cut costs**: IT infrastructure costs can become unwieldly if IT teams don't take the time to continuously examine a virtual infrastructure and its deliverables. Cost-cutting initiatives may range from replacing old servers and renegotiating vendor agreements to automating time-consuming server management tasks.

» **Prepare for failure**: Despite its failover hardware and high availability, even the most resilient virtual infrastructure can experience downtime. IT teams should prepare for worst-case scenarios by taking advantage of monitoring tools, purchasing extra hardware and relying on clusters to better manage host resources.

» A case study "Hypervisor management software".

Assignment

Objectives 33 from page no 258-161

Subjective Question 33 from page 161-162